Machine Learning in Data Streams - Project Results

Real-time CTR prediction with online feature selection

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Motivation

The essence of digital advertising

- The modern digital advertising market relies entirely on RTB systems.
- The final impression value (eCPM) depends heavily on estimating the click-through rate (CTR).
- Even a 0.1% improvement in prediction accuracy translates to multimillion-dollar savings [1].

Constraints [2]

- High velocity
- High dimensionality
- Constant retraining from scratch is too expensive. Models must update on the fly.

Challenge of non-stationarity

Concept drift [3]

Data distributions change over time. What a user liked yesterday might not be relevant today.

Feature drift [4]

The subset of features most relevant to the model changes dynamically.

The problem

Standard algorithms, like decision trees, fail to keep pace with these changes. This creates an **adaptation gap** – a period of significantly degraded performance while the model slowly learns the new distribution.

Proposed innovation

- Streaming random patches (SRP) [5]. One of the most effective ensemble solutions for evolving data streams.
- **Our novel extension:** integrating online feature selection (OFS) [6] into the SRP ensemble.
- **Mechanism:**
  - The algorithm evaluates feature usefulness on the fly. No static or random subspaces.
  - When drift is detected, newly created background learners receive access to the most recent and informative parameters.
- **Goal:** shorten recovery time ($t_{recovery}$) and reduce Log Loss during sudden market shifts.

Avazu dataset: EDA vol.1

Dataset characteristics

- 1,000,000 chronological instances to preserve natural drift.
- 24 features (mostly categorical).
- **Vivid cardinality:** `device_ip` has 313,002 unique values.
- **Class imbalance:** Only 16.02% of displays result in a click.

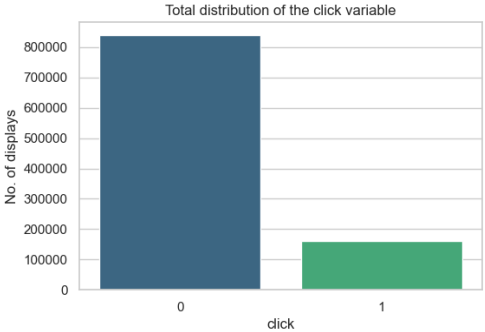


Figure: Total distribution of the `click` variable.

Avazu dataset: EDA vol.2

Time dependency

- The average CTR is not constant.
- Clear fluctuations depend on the hour of the day.
- This suggests the presence of natural CD in the stream.

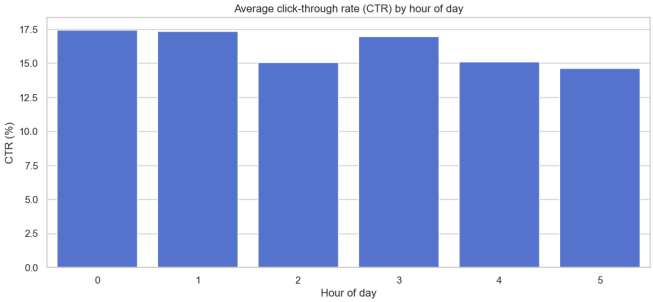


Figure: Average CTR by hour of day.

Avazu dataset: EDA vol.3

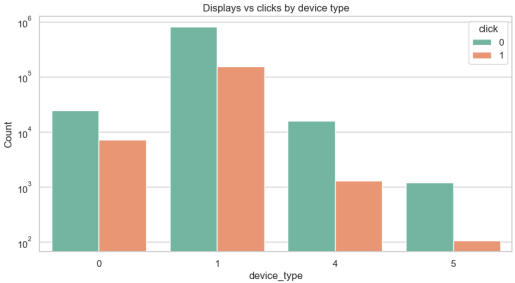


Figure: Displays vs clicks by device type.

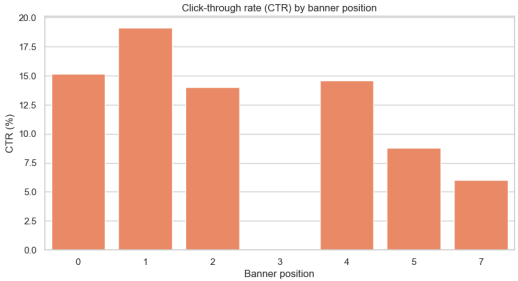


Figure: CTR by banner position.

- **Device type:** one platform dominates traffic, but propensity to click varies significantly.
- **Ad placement:** correlation between physical banner position and CTR.

Criteo dataset: EDA vol.1

Heterogeneous nature

- 40 attributes total. 13 continuous numerical (`I1-I13`) and 26 anonymized categorical (`C1 - C26`).
- Sample of 1,000,000 instances.
- **Class imbalance:** 25.49% clicks vs 74.51% non-clicks.

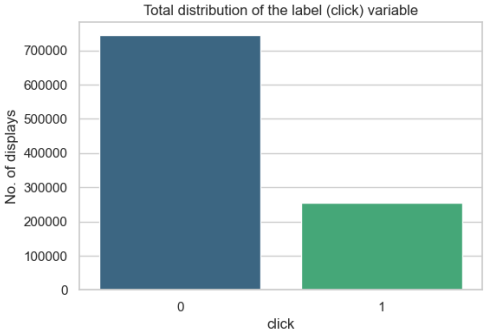


Figure: Total distribution of the `label` variable.

Criteo dataset: EDA vol.2

Data sparsity

- Presence of missing values (NaNs) across numerical columns (`I12` missing ∼75%).
- Requires zero-imputation.

Numerical skewness

- Heavily right-skewed, long-tail distributions.

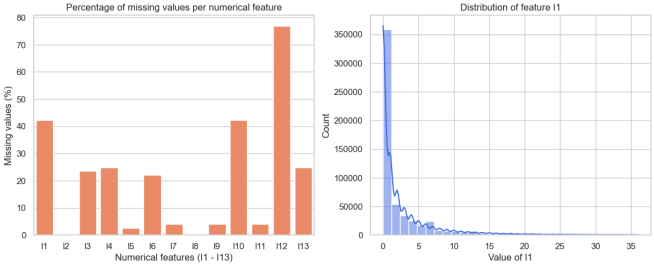


Figure: Missing values (left) and feature I1 skewness (right).

Feature engineering and hashing

Temporal and rolling window features

- **Avazu:** extracted specific time details and interaction features (`site_x_hour`). Implemented a sliding window ($W = 1000$).
- **Criteo:** Applied sliding window ($W = 1000$) to the first five numerical columns to compute moving averages, standard deviations, and deviations from the mean.

Dimensionality reduction via feature hashing [7]

- Used `FeatureHasher` to compress millions of categorical strings into a fixed-size vector of exactly 100 columns.
- Constant memory use, faster training times.

Synthetic stream: Agrawal generator [9]

Why synthetic data?

Measuring the exact speed of adaptation requires a 100% controlled environment.

- Simulates loan approvals (9 continuous attributes).
- Total length: 1,000,000 instances.

Abrupt drift injection (in MOA [8])

- **Instances 1 – 500,000:**
  function 1 (stable learning).
- **Instance 500,001:**
  sudden shift to function 2.
- **Instances 500,000 – 1M:**
  old rules become obsolete.

Empirical drift verification (Avazu)

ADWIN algorithm [10, 11]

- Verifying that CTR streams are truly non-stationary.
- Local CTR fluctuates significantly (14% to 18.5%).
- 4 distinct CD alerts triggered in 1M instances.

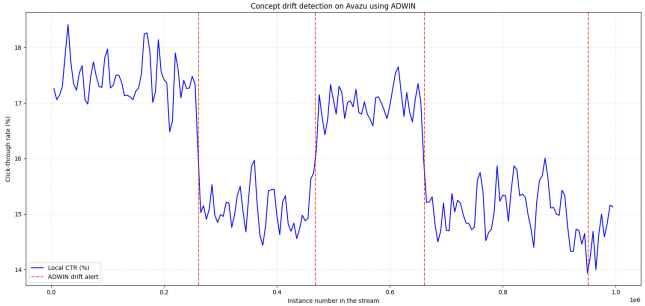


Figure: Concept drift detection on Avazu.

Empirical drift verification (Criteo)

Criteo volatility

- CTR rapidly climbs from initial values around 21% up to nearly 28%.
- 3 major drift alerts triggered.
- Natural drifts are prevalent, exposing the need for dynamic online learning models.

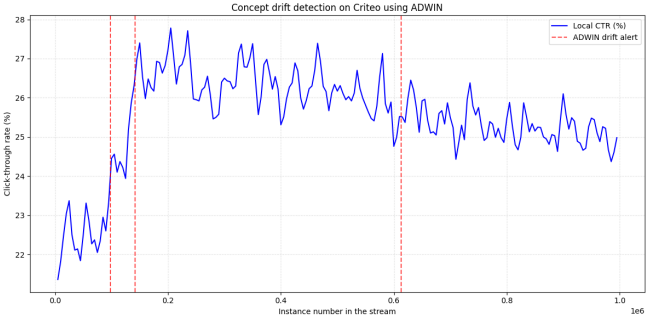


Figure: Concept drift detection on Criteo.

The adaptation gap

- **Baseline:** Hoeffding Adaptive Tree (HAT).
- Abrupt drift injected exactly at 500k.
- Error spikes abruptly to $> 30\%$.
- The subsequent slow recovery forms a distinct triangle of error.
- Standard trees are reactive and slow. OFS aims to flatten this curve and minimize $t_{recovery}$.

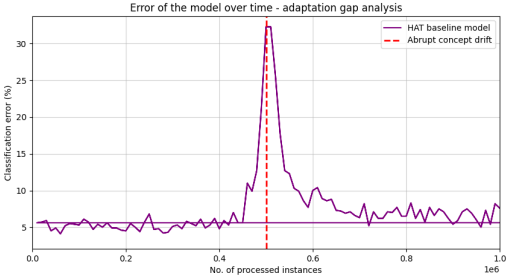
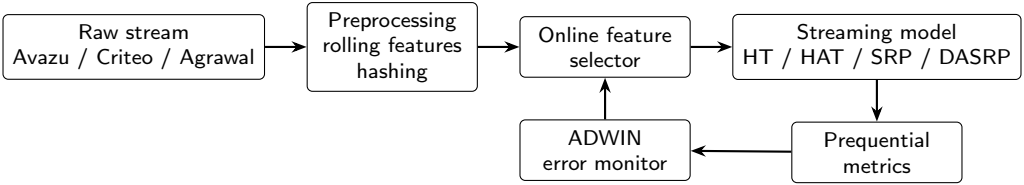


Figure: Error spike and recovery time ($t_{recovery}$).

Solution architecture: stream pipeline



Common path

Each instance is transformed once, filtered to the currently active feature subset, predicted, scored, and then used for training.

Feedback loop

ADWIN observes the online error stream. A drift alarm triggers feature re-ranking instead of waiting for a fixed retraining batch.

Solution architecture: drift-aware SRP

Adaptation logic

1. Keep a pre-drift window of recent instances.
2. After ADWIN fires, collect a post-drift window.
3. Recompute Information Gain for both windows.
4. Treat falling-IG features as weak and rising-IG features as strong.
5. Replace weak features inside affected ensemble subspaces.

Implementation details

- Ensemble: 10 Hoeffding-tree learners.
- Real CTR: `topK`=20, window=5,000, subspace size=20.
- Agrawal: `topK`=5, window=800, subspace size=4.
- Only learners whose subspace contains weak features are rebuilt; untouched learners keep their state.
- Rebuilt high-overlap learners start at weight 0.3 and recover by 0.001 per instance.

Prequential evaluation loop

Test-then-train protocol

For every stream instance, the model first predicts the label, then the prediction is scored, and only after that the same instance is used for training.

Why it matters

This avoids leakage from future observations and gives a time-resolved estimate of model quality under changing stream conditions.

1. Read next chronological instance.
2. Apply feature transformation and selection.
3. Predict click probability.
4. Update accuracy, log-loss and AUC.
5. Send error signal to ADWIN.
6. Train model on the same instance.

Implemented components

Data and models

- **Streams**: Avazu, Criteo, Agrawal.
- **Providers**: unified stream interface.
- **Models**: HT, HAT, SRP, DASRP.
- **Evaluation**: common prequential runner.

Selection and adaptation

- No feature selection.
- Static top-K after warm-up.
- Periodic online Information Gain ranking.
- Drift-aware reselection triggered by ADWIN.
- Selective subspace repair in DASRP.

Design goal

All providers, selectors and models expose common interfaces, so every configuration can be evaluated in the same loop without changing the evaluator.

Drift-aware feature selection

Before/after comparison

Instead of updating features on a fixed schedule, the selector waits for an ADWIN alarm and compares feature relevance before and after the detected drift.

$$\Delta IG_i = IG_{after,i} - IG_{before,i}$$

Decision rule

- Positive ΔIG means that a feature became stronger.
- Negative ΔIG means that a feature lost relevance.
- Strengthened features are added to the active set.
- Weakened features are removed from the active set.
- The final subset is corrected back to size K.

Main intuition

The selector reacts to changes in feature usefulness caused by drift, not only to absolute feature rankings at one point in time.

Model reaction to feature reselection

HT / HAT path

- Feature change creates a new input header.
- The whole model is reset.
- Previous tree structure is discarded.
- Learning starts again from an empty model.

DASRP path

- Each ensemble member owns its own subspace.
- Only subspaces containing weak features are changed.
- Only heavily affected learners are rebuilt.
- Reset learners vote with reduced weight.
- Their weight gradually returns to 1.0.

Key difference

Monolithic models lose all accumulated knowledge after reselection, while DASRP preserves unaffected learners and adapts only the damaged parts of the ensemble.

Evaluation procedure

Prequential evaluation (test-then-train)

- Cross-validation is inapplicable to endless streams.
- Each incoming instance is first used to test the model, and only then passed to train the model.
- Provides evaluation on unseen data.

Chosen metrics

- **Logarithmic loss (Log Loss) [12]:** heavily penalizes confident yet incorrect predictions.
- **AUC ROC [13]:** ability to correctly rank positive vs negative classes.

Experiment scale in this repository

48 experiments: 36 baseline combinations plus 12 drift-aware variants. Each dataset was evaluated for 200,000 instances, with metric snapshots every 1,000 instances.

Final evaluation: real CTR streams

Dataset	Method	Accuracy	Log Loss	AUC
Avazu	SRP + static top-K	0.8243	0.4659	0.7022
Avazu	**DASRP + drift-aware SRP**	**0.8253**	**0.4332**	**0.7048**
Criteo	SRP + online ranking	0.7557	0.5522	0.6512
Criteo	**DASRP + drift-aware SRP**	**0.7583**	**0.5291**	0.6511
Criteo	SRP + static top-K	**0.7634**	0.5622	**0.6754**

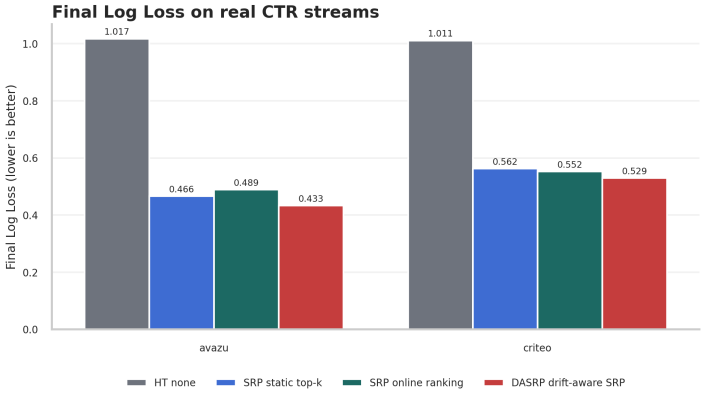
Main gain

DASRP achieved the lowest Log Loss on both real CTR streams: 7.0% lower than the best Avazu baseline and 4.2% lower than the best Criteo Log Loss baseline.

Important nuance

On Criteo, static SRP still produced the best AUC and accuracy. The proposed method mainly improved probability calibration.

Real CTR: final Log Loss visualized



Main message

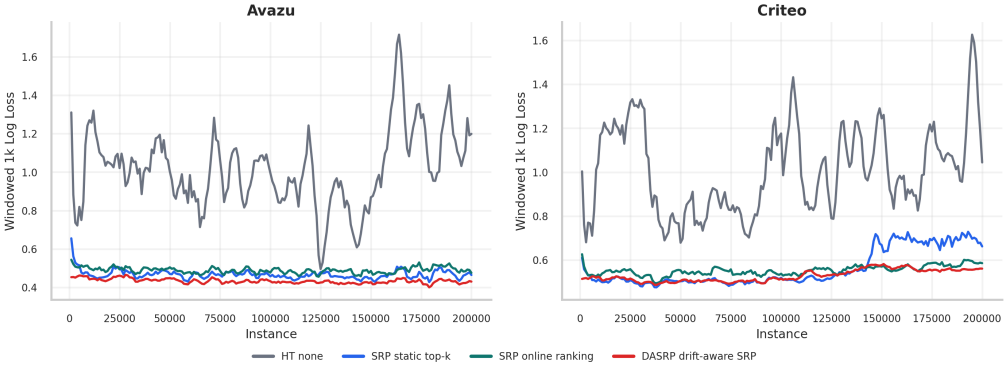
- DASRP obtains the lowest final Log Loss on both real CTR datasets.
- Avazu: 0.433 vs 0.466 for the best SRP baseline.
- Criteo: 0.529 vs 0.552 for the best Log Loss baseline.

Interpretation

The strongest improvement is probability calibration, not only class accuracy.

Evaluation dynamics: real CTR streams

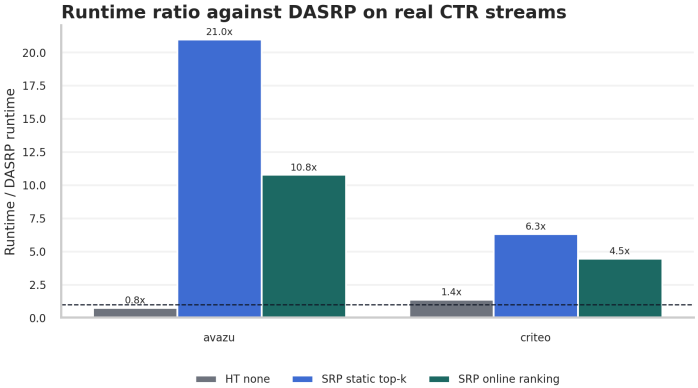
Probability loss over time on real CTR streams



What the curves show

DASRP stays among the lowest-loss methods throughout the stream. The gain is especially visible on Avazu, while Criteo shows a later degradation of static SRP that DASRP avoids.

Runtime comparison



Why it matters

- Real CTR experiments are high-dimensional after hashing and rolling features.
- On Avazu, SRP static top-K took about 21x the DASRP runtime.
- HT is faster but less competitive in final Log Loss.

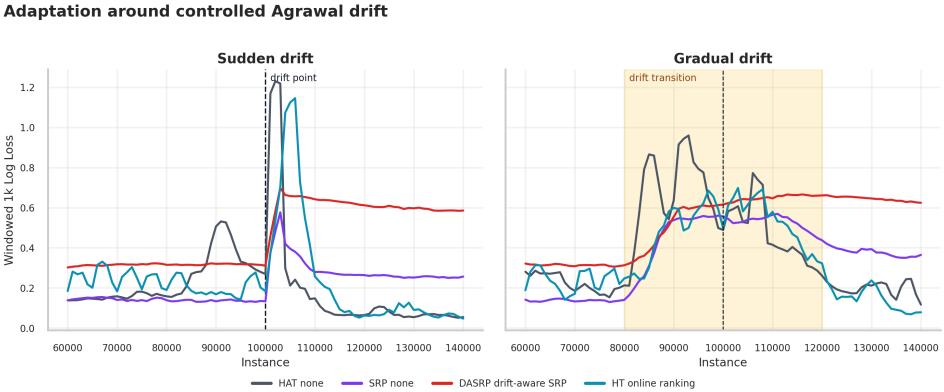
Final evaluation: synthetic drift

Dataset	Method	Accuracy	Log Loss	AUC
Sudden Agrawal	HAT + no selector	**0.9559**	**0.1604**	0.9966
Sudden Agrawal	HT + online ranking	0.9279	0.1939	**0.9982**
Sudden Agrawal	DASRP + drift-aware SRP	0.8435	0.4553	0.8838
Gradual Agrawal	HT + online ranking	0.9195	**0.2171**	0.9979
Gradual Agrawal	HAT + no selector	**0.9282**	0.2420	0.9975
Gradual Agrawal	DASRP + drift-aware SRP	0.8029	0.4848	0.9067

Interpretation

In the low-dimensional Agrawal setting, standard adaptive trees and periodic online ranking already recover very well. Drift-aware selection generated many extra alarms, so the proposed mechanism needs stricter tuning for simple synthetic streams.

Synthetic drift: adaptation curves



Interpretation

Standard adaptive trees and periodic online ranking recover quickly on low-dimensional Agrawal streams. DASRP is more conservative here, so its drift thresholds need separate tuning for simple synthetic data.

Conclusion

What worked

- Feature hashing and rolling features gave a fixed-memory CTR stream representation.
- Drift-triggered feature adaptation improved real-stream Log Loss.
- DASRP kept unaffected learners alive instead of resetting the whole ensemble.

What remains

- Tune ADWIN delta, window size, and IG threshold per data regime.
- Quantify recovery time and area under the loss spike around drift events.
- Test larger ensembles and full 1M-instance CTR runs.

Takeaway

Online feature selection is most useful when the stream is high-dimensional and feature relevance drifts. In this project, the strongest evidence is on Avazu and Criteo, where DASRP delivers the best probability-aware loss.

References I

[1] H Brendan McMahan, Gary Holt, David Sculley, Michael Young, Dietmar Ebner, Julian Grady, Lan Nie, Todd Phillips, Eugene Davydov, Daniel Golovin, et al.
Ad click prediction: a view from the trenches.
In *Proceedings of the 19th ACM SIGKDD international conference on Knowledge discovery and data mining*, pages 1222–1230, 2013.

[2] Pedro Domingos and Geoff Hulten.
Mining high-speed data streams.
In *Proceedings of the sixth ACM SIGKDD international conference on Knowledge discovery and data mining*, pages 71–80, 2000.

[3] João Gama, Indrė Žliobaitė, Albert Bifet, Mykola Pechenizkiy, and Abdelhamid Bouchachia.
A survey on concept drift adaptation.
ACM computing surveys (CSUR), 46(4):1–37, 2014.

References II

[4] Haiguang Li, Xindong Wu, Zhao Li, and Wei Ding.
Group feature selection with streaming features.
In *2013 IEEE 13th International Conference on Data Mining*, pages 1109–1114. IEEE, 2013.

[5] Heitor Murilo Gomes, Jesse Read, and Albert Bifet.
Streaming random patches for evolving data stream classification.
In *2019 IEEE international conference on data mining (ICDM)*, pages 240–249. IEEE, 2019.

[6] Congcong Liu, Yuejiang Li, Fei Teng, Xiwei Zhao, Changping Peng, Zhangang Lin, Jinghe Hu, and Jingping Shao.
On the adaptation to concept drift for ctr prediction.
arXiv preprint arXiv:2204.05101, 2022.

References III

[7] Kilian Weinberger, Anirban Dasgupta, John Langford, Alex Smola, and Josh Attenberg.
Feature hashing for large scale multitask learning.
In *Proceedings of the 26th annual international conference on machine learning*, pages 1113–1120, 2009.

[8] Albert Bifet, Geoff Holmes, Bernhard Pfahringer, Philipp Kranen, Hardy Kremer, Timm Jansen, and Thomas Seidl.
Moa: Massive online analysis, a framework for stream classification and clustering.
In *Proceedings of the first workshop on applications of pattern analysis*, pages 44–50. PMLR, 2010.

[9] Rakesh Agrawal, Tomasz Imielinski, and Arun Swami.
Database mining: A performance perspective.
IEEE transactions on knowledge and data engineering, 5(6):914–925, 1993.

References IV

[10] Albert Bifet and Ricard Gavalda.
Learning from time-changing data with adaptive windowing.
In *Proceedings of the 2007 SIAM international conference on data mining*, pages 443–448. SIAM, 2007.

[11] Albert Bifet and Ricard Gavalda.
Adaptive learning from evolving data streams.
In *International symposium on intelligent data analysis*, pages 249–260. Springer, 2009.

[12] Vladimir Vovk.
The fundamental nature of the log loss function.
In *Fields of logic and computation II: Essays dedicated To Yuri Gurevich on the Occasion of His 75th Birthday*, pages 307–318. Springer, 2015.

References V

[13] Tom Fawcett.
An introduction to roc analysis.
Pattern recognition letters, 27(8):861–874, 2006.

Thank You for Your Attention!

Feel free to ask questions :)